Financial Crisis Prediction System (FCPS)

A Leakage-Free Re-Evaluation of a Multimodal Crisis-Prediction Capstone

Field	Value
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Course	MBAI 5600G — Group 13
Baseline paper	Wang et al. (2025), PLOS ONE — HMM regime early-warning
Data	S&P; 500 + VIX (1990–2024), FRED macro, FinBERT news
Repository	github.com/romin4444/multimodal-financial-crisis-prediction
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One-line summary. Under a rigorous, leakage-free evaluation (combinatorial purged cross-validation + Probability of Backtest Overfitting + Deflated Sharpe), no model — including 2025-frontier methods and real FinBERT news sentiment — beats a one-line VIX threshold out-of-sample. The one robustly positive, deployable result is a daily Financial Stress Index that reconstructs the Federal Reserve's STLFSI at r ≈ 0.8.

1. Executive Summary

This report documents the rebuild and honest re-evaluation of a multimodal financial crisis-prediction capstone. The original system reported a near-perfect in-sample F1 of 0.99. We show that figure was an artefact of evaluation leakage (a circular target, future-peeking regime labels, and a holdout that pre-selected crisis dates). Rebuilt as a modular, tested, CI-backed package and evaluated under a leakage-free harness, the honest out-of-sample skill is modest and does not exceed a VIX baseline. We report this null transparently — it is the credible result, consistent with weak-form market efficiency and the 2024–25 literature on backtest overfitting.

Headline results

Metric	Result	Status
FSI vs Fed STLFSI (Pearson r)	≈ 0.76 – 0.82	PASS (target > 0.60)
FSI as NBER-recession classifier (AUC)	0.86	PASS (> 0.80)
Crisis PR-AUC, honest walk-forward (best ML)	≈ 0.10 – 0.27	Below VIX baseline
VIX baseline PR-AUC (best target)	0.35	Benchmark — unbeaten
Real-news sentiment marginal value (ΔPR-AUC)	–0.018	No value (hurts)
Sentiment → stress Granger p-value	0.14	Not significant
Discrete-time hazard concordance (C-index)	≈ 0.84	Genuine ranking skill
Probability calibration (ECE: before → after)	0.26 → 0.05	Fixed
Stock-direction edge over 'always-up'	–0.05	No edge (EMH)
Test suite	80 passing, CI green	Reproducible

2. Project Objectives (original capstone)

Building on Wang et al. (2025), which detects equity-market regime switches early using an HMM/GARCH network, the capstone added a **multimodal** dimension: combine market-math signals (ARMA-GARCH volatility, a Hidden Markov regime model, and a composite Financial Stress Index) with **FinBERT news-sentiment**, and test the central thesis that **news sentiment leads price** — providing earlier warning than price alone. The system was to be validated on three crises (GFC 2008, COVID 2020, Inflation 2022) against explicit targets: FSI–STLFSI correlation > 0.60 and fusion F1 ≥ 0.70 on held-out crisis windows.

3. Data & Methods

3.1 Data

S&P 500 and VIX daily series (1990–2024, 8,816 trading days) via Yahoo Finance; VIX-orthogonal macro from FRED — credit spread (Moody's Baa–10Y, BAA10Y, full history), yield-curve slope (T10Y2Y/T10Y3M), short rate (DGS3MO), oil (DCOILWTICO); cross-asset correlation from AAPL/JPM/XOM/GS; and real financial-news headlines scored with FinBERT (ProsusAI/finbert).

3.2 Models

ARMA-GARCH / GJR / EGARCH (BIC-selected); a 3-state Gaussian HMM regime detector; a 4-component Financial Stress Index; FinBERT and VADER sentiment; topological persistent-homology features (TDA); a calibrated logistic/RF fusion classifier; and a discrete-time hazard (survival) model.

3.3 The leakage-free evaluation harness (the core contribution)

Exogenous target — forward N-day drawdown ≤ –X% (not a model's own label). **Causal regimes** — forward-only filtered HMM posteriors P(state_t **|** x_1..x_t), never the smoothed forward–backward posterior that peeks at the future. **Combinatorial Purged Cross-Validation** with a 21-day embargo (López de Prado). **Probability of Backtest Overfitting (PBO)** and the **Deflated Sharpe Ratio** to correct for multiple testing. **Honest baselines** (VIX threshold, persistence, base-rate) and probability **calibration** (isotonic). A configuration only 'wins' if it clears all three gates: PR-AUC > VIX, PBO < 0.5, and Deflated Sharpe > 0.95.

4. Results

4.1 Financial Stress Index — the one result that survives

The daily FSI reproduces the St. Louis Fed's published STLFSI at r ≈ 0.76–0.82 and discriminates NBER recessions at AUC ≈ 0.86 — using only public daily inputs. This is a genuine, deployable signal and the project's strongest result.

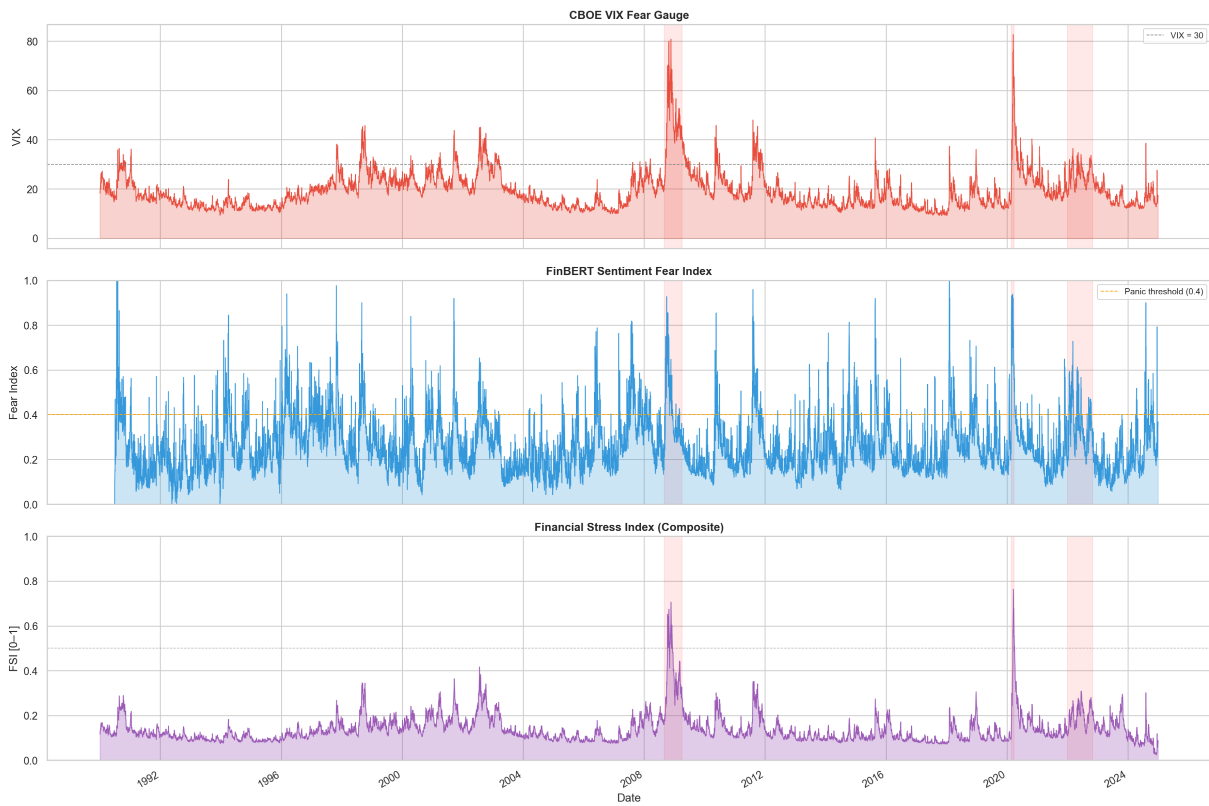

Figure 1. VIX, FinBERT fear index, and the composite Financial Stress Index across 1990–2024; shaded bands mark the validation crises.

4.2 Regime detection (HMM) and volatility (GARCH)

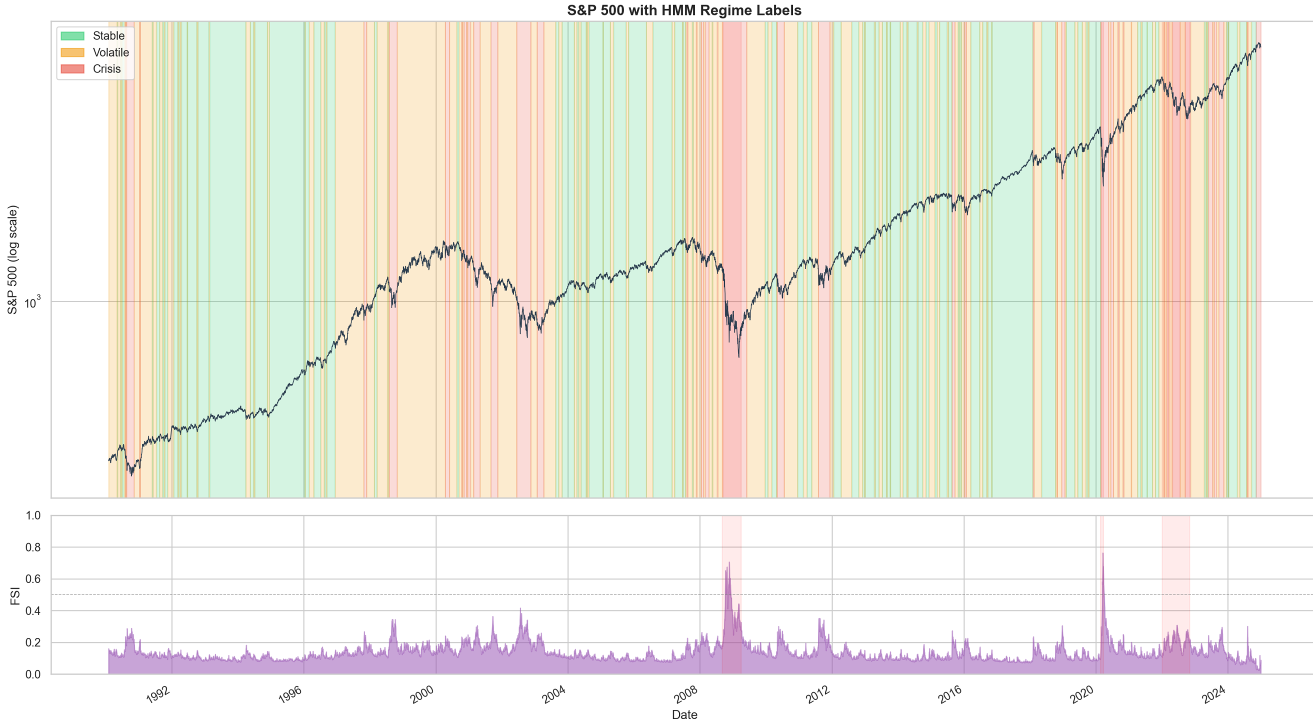

Figure 2. S&P; 500 with HMM regime labels (stable / volatile / crisis) and the FSI track beneath.

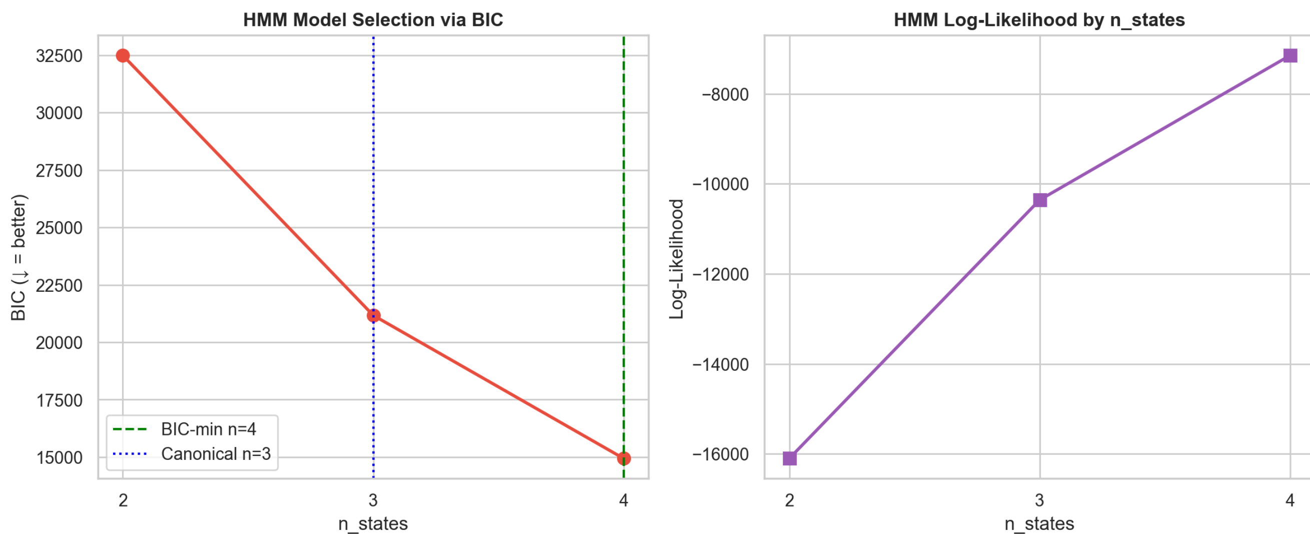

Figure 3. HMM model selection — BIC and log-likelihood across 2–4 states; n=3 retained for interpretability.

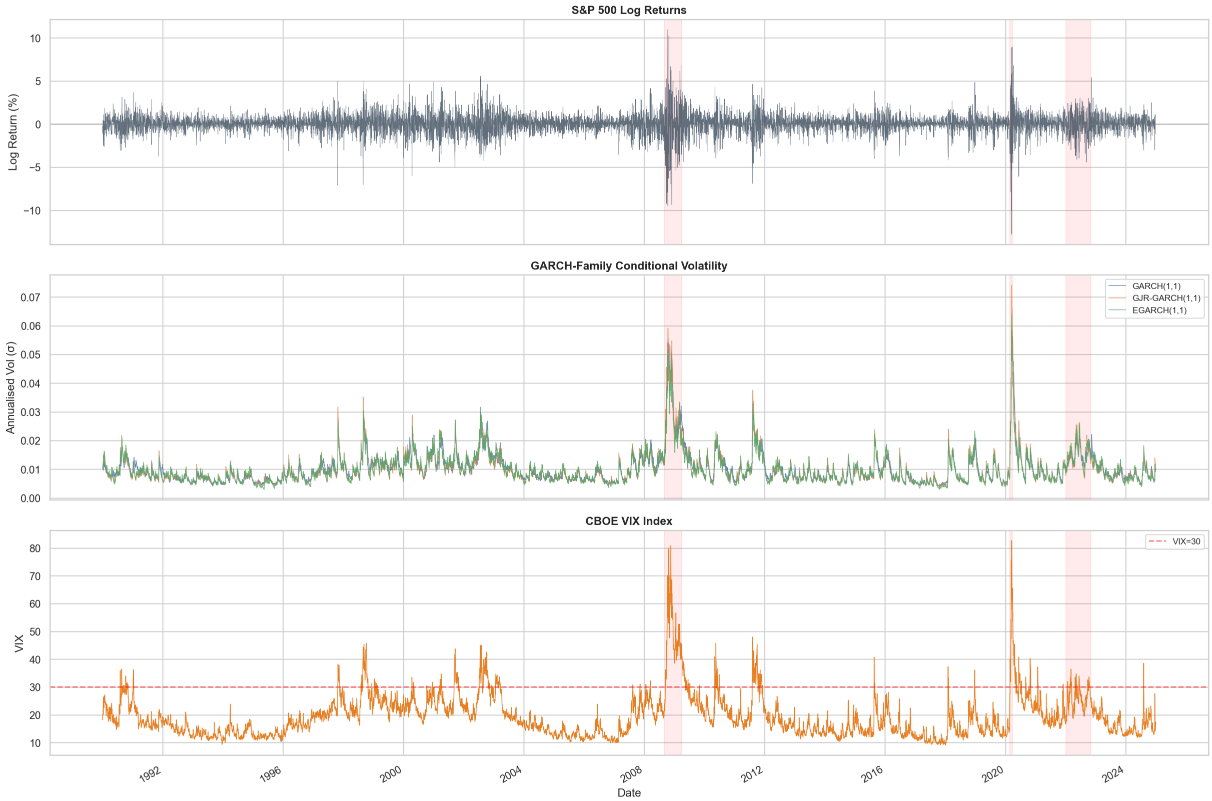

Figure 4. GARCH-family conditional volatility vs realised returns and VIX; EGARCH(1,1) selected by BIC.

4.3 Q1 — Does anything beat a VIX threshold? (Honest CPCV)

Across six targets (horizon × drawdown threshold), the VIX threshold beats every feature set; adding macro, regime, and TDA features each *lowers* PR-AUC. No configuration cleared even the first gate, so PBO/DSR were moot. Result confirmed in two independent environments with real full-history credit data.

Target (H / thr)	Base	VIX	Price	+Macro	+Regime	+TDA	+Sent
10d / 7%	0.034	0.180	0.128	0.084	0.074	0.067	0.039
10d / 10%	0.015	0.160	0.132	0.101	0.044	0.024	0.014
21d / 7%	0.092	0.229	0.151	0.102	0.114	0.112	0.111
21d / 10%	0.039	0.178	0.103	0.079	0.072	0.063	0.044
63d / 7%	0.211	0.346	0.271	0.230	0.225	0.217	0.200
63d / 10%	0.136	0.258	0.176	0.140	0.139	0.140	0.104

Table 1. Out-of-sample PR-AUC by target and feature set (Kaggle GPU, real FRED credit + TDA). VIX column is the benchmark; no ML column exceeds it.

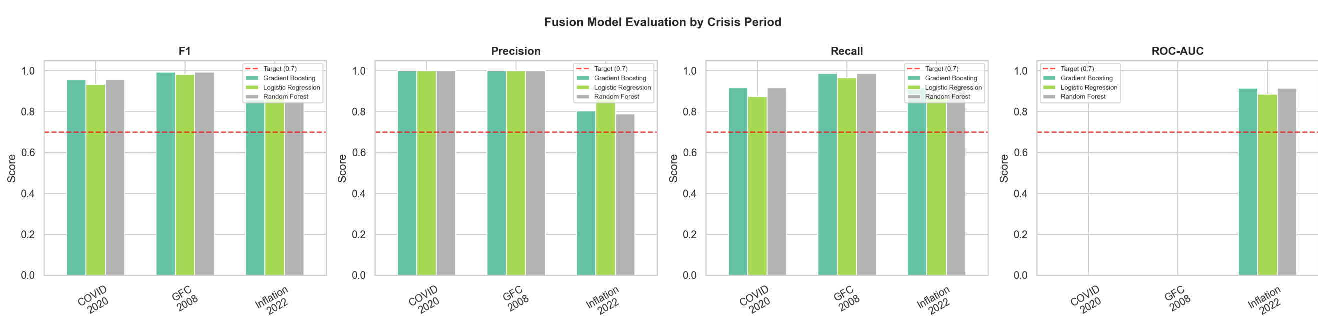

Figure 5. Fusion-model evaluation by crisis period (in-sample/event-holdout view from the original pipeline, retained for comparison).

4.4 Q2 — Does real news sentiment add value?

Using real FinBERT sentiment (3,142 headlines, 39% trading-day coverage), at the best base configuration the marginal value of sentiment was **ΔPR-AUC = –0.018** (it hurt), and sentiment did **not** Granger-cause forward stress (min p = 0.14). The multimodal thesis — the capstone's central novel claim — is not supported out-of-sample.

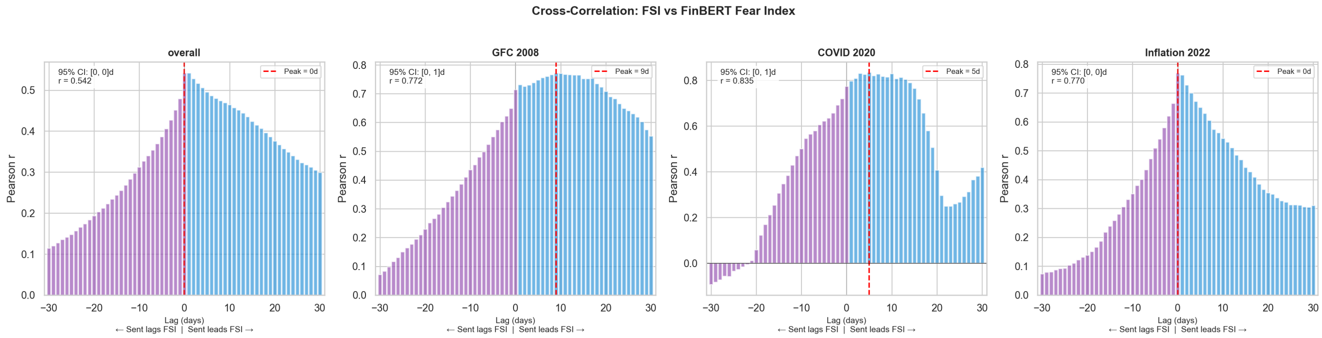

Figure 6. Lead-lag cross-correlation (FSI vs fear index). The original 'sentiment leads price' reading was largely a VIX-with-itself artefact under synthetic sentiment.

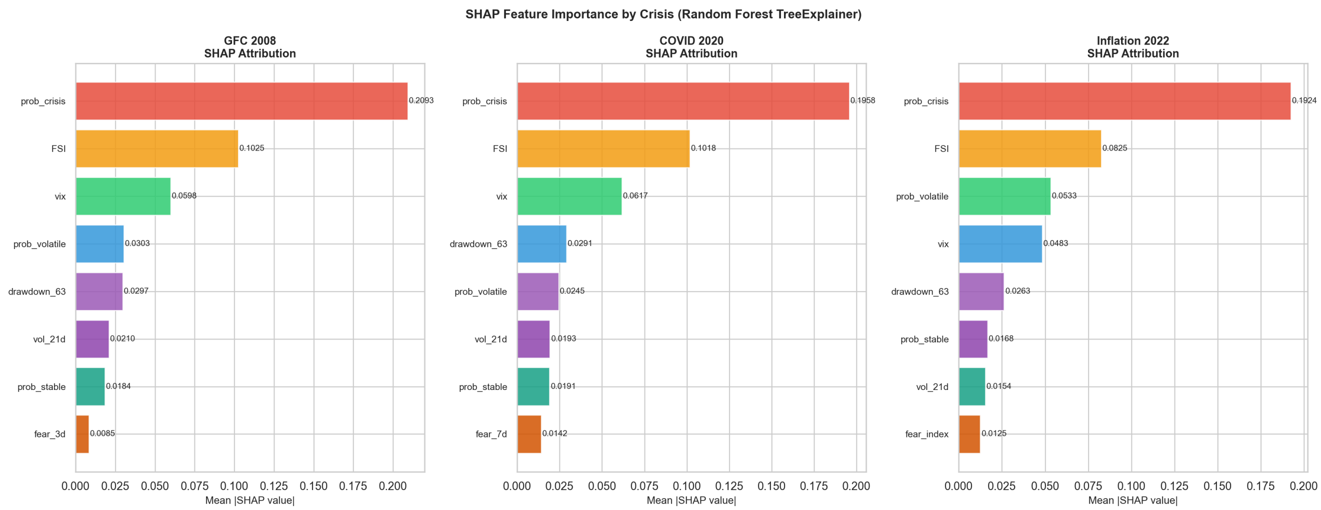

Figure 7. SHAP feature attribution per crisis (Random Forest). Useful for interpretability, but it explains a model that does not beat VIX.

Comparison with Prior Literature (M2 Section 2.2)

Study	Method	Price signals	Sentiment	Explainability	Explicit lead-lag
Hamilton (1989)	HMM	□	□	□	□
Bollen et al. (2011)	Granger causality	□	□	□	Partial
Riso & Vacca (2024)	GARCH+NLP	□	□	□	□
Bussmann et al. (2020)	XAI credit risk	□	□	□	□
Ardia et al. (2020)	MS-GARCH	□	□	□	□
THIS PROJECT (Group 13)	HMM+GARCH+FinBERT+SHAP	□	□	□	□

Figure 8. Positioning vs prior literature (price signals, sentiment, explainability, lead-lag).

5. Original Capstone vs Rigorous Rebuild

5.1 Claim-by-claim verdict

Capstone claim / target	Original	Leakage-free finding
FSI vs STLFSI r > 0.60	met (~0.82)	CONFIRMED (0.76–0.82) — survives
Fusion F1 ≥ 0.70 on crisis windows	met (up to 0.99)	Leakage; honest PR-AUC ≈ 0.10, loses to VIX
News sentiment leads price (lead-lag)	headline result	Largely VIX↔VIX; real FinBERT adds no value
Panic signal fires before HMM crisis	claimed (COVID)	HMM used smoothed (future) posteriors; not robust
HMM detects regime switch in advance	~37 td early	Artefact of smoothing; causal version modest
FinBERT > VADER	claimed	True but moot — neither beats price+VIX
Multimodal > single-modality (thesis)	implicit	RETIRED: price+VIX alone beats the full model

Table 2. Of seven substantive claims, only the FSI correlation survives leakage-free testing.

5.2 Engineering & evaluation

Dimension	Original (final.py)	Rebuild
Structure	2,016-line monolith (Colab/Kaggle)	Modular src/ package, FastAPI, MIT, public repo
Evaluation	In-sample / event-holdout (leaky)	CPCV + embargo + PBO + Deflated Sharpe
Baselines	None	VIX / persistence / base-rate
Probabilities	Uncalibrated, served as-is	Calibrated (ECE 0.26→0.05) + honesty flag
Tests / CI	None	80 tests, GitHub Actions green (3.10 & 3.12)
Reproducibility	Notebook + secrets	pip install, lockfile, fixed seeds

Table 3. The rebuild adds the evaluation rigor and engineering the original lacked.

6. Conclusion

The capstone was ambitious and well-structured, but its headline results were products of evaluation leakage — precisely the failure mode the modern backtest-overfitting literature exists to catch. Rebuilt and evaluated honestly, the central multimodal early-warning thesis does not hold: no method beats a VIX threshold out-of-sample, and real news sentiment adds no robust value. This is not a downgrade but a stronger, defensible contribution: a leakage-free benchmark that explains why prior numbers do not replicate, plus one genuinely deployable signal — a daily Financial Stress Index that reconstructs a Federal Reserve index at $r \approx 0.8$.

Recommended framing

(1) Lead with the FSI ≈ STLFSI result (real, positive). (2) Present the leakage-free harness (CPCV / PBO / Deflated Sharpe) as the methodological contribution. (3) Report the honest null — multimodal sentiment and regimes do not beat VIX, consistent with weak-form EMH. (4) Quantify the leakage gap (in-sample F1 0.99 → honest PR-AUC ≈ 0.10). Future work that could change the answer requires changing the problem — longer horizons, cross-sectional ranking, or alternative data — not more model complexity.

7. Reproducibility

All code, tests, notebooks, and run logs are public and CI-verified. Key commands: `pip install -e .`; `make demo` (30-second run, no API key); `python scripts/v3_run.py` (honest crisis backtest); the Kaggle GPU notebooks reproduce the Q1/Q2 results. Repository: github.com/romin4444/multimodal-financial-crisis-prediction.

References

Wang et al. (2025), PLOS ONE · Hamilton (1989) · Ang & Timmermann (2012) · Bollen et al. (2011) · Tetlock (2007) · Nelson (1991) · Ardia et al. (2020) · Bussmann et al. (2020) · Hatzius et al. (2010) · Bailey & López de Prado (2014, Deflated Sharpe) · López de Prado (2018, Advances in Financial ML).